

KAVINDA KEHELELLA

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RESEARCH INTERESTS

My research focuses on Machine Learning, Natural Language Processing, and Artificial Intelligence, spanning both algorithmic foundations and practical applications. My work includes LLM consistency and explainability, knowledge distillation, multimodal AI for material discovery and chemistry, and machine learning for neural signal processing. I seek to bridge methodological advances with practical validation through interdisciplinary collaborations and large-scale experiments on real-world data.

Keywords: Machine Learning, LLM Robustness & Consistency, Knowledge Distillation, Natural Language Processing, Explainable AI, Mechanistic Interpretability, Multimodal AI for Material Science, Graph Neural Networks, Neural Decoding, Digital Signal Processing, Wireless Sensing

EDUCATION

University of Maryland, College Park, MD Started Aug. 2023
Ph.D. in Electrical and Computer Engineering (CGPA: 3.92/4.00)
Advised by Prof. Sanghamitra Dutta

University of Moratuwa, Sri Lanka Graduated May 2022
B.Sc. (Hons.), Electronic and Telecommunication Engineering, First Class (CGPA: 3.99/4.2)
Advised by Dr. Chamira Edussooriya, Dr. Ranga Rodrigo
Thesis: Radio frequency wireless sensing using micro-doppler signatures

WORK EXPERIENCE

Software Engineer May 2022 – Aug. 2023
MyBudget Holdings (Pvt) Ltd., Colombo, Sri Lanka
Designed and developed a customer web portal using Go and Java, integrated PostgreSQL and Kafka backend systems, and maintained technical documentation using Confluence.

Intern Software Engineer Oct. 2020 – Apr. 2021
Axiata Digital Labs, Colombo, Sri Lanka
Developed backend services using Spring Boot, implemented unit tests, and contributed to software development within an Agile team.

PUBLICATIONS

- **K. Kehelella, S. Dutta**, “COFFEQA: Counterfactual-Guided Few-Shot Distillation for LLM Question Answering,” Under Review at NeurIPS 2026.
- **K. Kehelella, X. Wu, H. Ali, C. Yang, J. Xie, Y. Lee, R. Chen, S. Bhattacharyya** “RNDC-Lab: A Real-Time Calcium Imaging-Based Adaptive Neural Decoding Framework,” Accepted for publication in Frontiers in Neuroscience, 2026.
- **K. Kehelella, G. Leelarathne, D. Marasinghe, N. Kariyawasam, V. Ariyaratna, A. Madanayake, R. Rodrigo, C. US Edussooriya**, “Vision Transformer with Convolutional Encoder-Decoder for Hand Gesture Recognition using 24 GHz Doppler Radar,” IEEE Sensors Letters, 2022.

SKILLS AND TOOLS

Programming: Python, C/C++, MATLAB, Java, Go, SQL

ML Frameworks: PyTorch, TensorFlow/Keras, scikit-learn

LLM Research: Huggingface Transformers, T5, Qwen, Gemma, ChemBERTa, MatSciBERT, Knowledge Distillation, Mechanistic Interpretability

ML Methods: Graph Neural Networks (GNNs), Multimodal Learning, Computer Vision, Image processing, Signal Processing

Tools: Git, Docker, Jupyter, PostgreSQL, MySQL, Kafka, JIRA, Confluence

RESEARCH EXPERIENCE

Counterfactual-Guided Few-Shot Distillation for LLM Question Answering 2026

Developed a few-shot knowledge distillation framework that incorporates factual examples with counterfactuals to improve knowledge transfer in LLMs for open-domain question answering. Introduced the Context Following Distillation (CFD) metric to validate generated counterfactuals based on teacher context dependence and answer agreement. Evaluated COFFEQA across eight QA benchmarks using T5 and Qwen2.5 teacher-student architectures, including analyses of teacher-student agreement and distillation near the teacher's decision boundary.

Improving Consistency in LLM Question Answering Ongoing

Investigated why LLMs produce inconsistent outputs on semantically equivalent, paraphrased questions by analyzing hidden-state representations layer-by-layer using Gemma-2B-it on HotpotQA. Trained probing classifiers to locate the attention heads most responsible for paraphrase-induced inconsistency, then applied targeted activation-editing interventions at those heads to reduce inconsistency. Built an evaluation pipeline reporting per-head and per-layer inconsistency deltas before and after editing, with aggregate improvement rates across evaluation instances.

A Unified Multimodal GNN-LLM Framework for Material Property Prediction Ongoing

Developed a multimodal learning framework for biopolymer property prediction by combining molecular graph representations with semantic representations from domain-specific language models, including ChemBERTa-2 and MatSciBERT. Engineered a substructure-to-text generation pipeline to encode per-component chemistry and inter-component interactions into natural-language descriptions. Integrated structural and semantic embeddings to improve downstream property prediction over a GNN-only baseline. Currently exploring an AI agent system to improve reasoning over material compositions and properties.

RNDC-Lab: Real-Time Calcium Imaging-Based Adaptive Neural Decoding 2025

Developing a real-time, end-to-end neural decoding system by integrating two main modules, NDSEP and NeuroGRS, to improve suitability for real-time neural decoding. Tested the system on both simulated and real-world two-photon calcium imaging datasets.

Dynamically Reconfigurable Perception via Dataflow Parameterization 2024

Developed a parameterized method for pruning a large deep learning model for deployment in resource-constrained environments. The model is pruned during inference. Conducted experiments using ResNet50 with a Convolutional Block Attention Module on the ImageNet dataset.

Sentiment Transition Analysis in Updated Amazon Reviews 2024

Extracted and analyzed updated Amazon reviews across five product categories, comparing sentiment analysis approaches (GPT-4, a BERT-based sentiment classifier, VADER, and a fine-tuned DistilBERT) and highlighting the strengths and limitations of each.

Radio Frequency Wireless Sensing using Micro-Doppler Signatures 2022

Developed a Doppler radar system with directional sensing capability operating in the 2.45 GHz Industrial, Scientific, and Medical (ISM) band to recognize hand gestures in real time using machine learning, achieving more than 94% accuracy. The system includes the design and implementation of a patch antenna as the transmitter, an array of four patch antennas as the receiver, an analog multi-beam beamforming network with a Butler matrix, and backend software to recognize seven hand gestures. The proposed architecture consists of three modules: a convolutional encoder-decoder, an attention module with three transformer layers, and a multi-layer perceptron for real-time gesture recognition.

OTHER PROJECTS

Multi-rate Tx/Rx system 2023

Implemented multi-rate communication system using MATLAB to recover a signal in presence of noise.

Diversity Combining Scheme Comparison 2022

Analyzed the PDFs and CDFs of Rayleigh and Rician-distributed random variables. Compared diversity combining schemes in independent and identically distributed fading channels, including Maximal Ratio Combining (MRC) and Switch-and-Stay Combining (SSC).

- Lossless Data Compression** 2021
Implemented lossless data compression algorithms, including Huffman coding, arithmetic coding, and dictionary coding. Implemented the Lempel-Ziv-Welch (LZW) algorithm as the dictionary coding method.
- Human Voice Amplifier with Background Noise Reduction** 2020
Designed a Class AB power amplifier with a low-pass filter to amplify human voice signals while reducing background noise, applying concepts from power electronics.
- Eye Diagram Generation for PAM Signaling** 2019
Generated eye diagrams to analyze inter-symbol interference (ISI), sampling timing accuracy, and noise immunity. Analyzed the robustness of Binary Phase Shift Keying (BPSK) and 4-PAM (Pulse Amplitude Modulation) digital communication systems using eye diagrams.
- FIR Bandstop Filter Design** 2019
Designed an FIR bandstop filter using the Fourier transform and a Kaiser window. Applied the FFT algorithm to obtain the frequency response and the DFT to evaluate filter characteristics.

TEACHING EXPERIENCE

- Graduate Teaching Assistant** Aug. 2023 – Present
Dept. of Electrical and Computer Engineering, University of Maryland
ENEE440 Microprocessors; ENEE245 Digital Circuits and Systems Laboratory; ENEE408C/M Capstone Design Projects
- Visiting Teaching Assistant** Jul. 2022 – Jul. 2023
Dept. of Electronic & Telecom. Engineering, University of Moratuwa
EN1094 Laboratory Practice; EN4720 Cyber Physical Systems

HONORS AND AWARDS

- Graduate School Dean's Fellowship, University of Maryland (2023)
- Finalist, Hackstat 2.0 – ML/Data Science Competition, University of Colombo (2019)
- Finalist, SLIIT Datathon – Data Science Competition (2019)
- Finalist, Datastorm 1.0 – ML/Data Science Competition, University of Moratuwa (2019)
- Ranked 243rd (World), IEEE Xtreme 14.0 (2020)
- Ranked 489th (World), IEEE Xtreme 13.0 (2019)
- Mahapola Merit Scholarship, Government of Sri Lanka (2017)